

Chapter 3

A Brief History of Site Catchment Analysis

“It therefore seems (regrettably) necessary to coin an independent name for the study of the relationships between technology, and those natural resources lying within economic range of individual sites. We propose the term ‘site catchment analysis’.”

– Vita-Finzi and Higgs 1970, 5

3.1 Introduction

Chapters 1 and 2 introduced the topic of research being examined in this thesis, provided a brief description of the southern Levant, and outlined some of the more pertinent archaeological background information. Before presenting the methodology used in this research, a brief overview of the development (and methods) of site catchment analysis will be given in this chapter. The purpose of this overview is to provide the reader with a background with which they can compare the methodologies developed and presented in this research with those undertaken by previous researchers. Coupled with the discussion of site catchment analysis will be a discussion of Geographic Information Systems and predictive modelling, and the roles which they have played in the history of archaeological analysis.

Once this background information has been presented, the methodology will be presented in detail in Chapter 4 (page 47). As well, it provides a context for a further extension of this methodology presented in Chapter 6 (page 325). This shows how catchment analysis can be effectively integrated into an analysis of networks on a regional level.

Innovations in technology and its spread in society can be divided into four phases (Foote and Lynch 2000):

Initiation Innovation makes something available for the first time.

Contagion Experimentation follows exploring how the innovation can be changed to accommodate the needs of research/commerce (it is not necessary for all of these experiments to work, but at least some must.)

Co-ordination The best applications of the innovation gradually gain acceptance and are developed collaboratively; experiment co-ordination results in a distribution of any costs of development and implementation.

Integration The innovation gains widespread acceptance and is thus integrated into mainstream research.

Technology has played an important role in advancements of archaeological investigations. In the following discussion, it can be seen how the technological advancements relating to site catchment analysis can, for the most part, be seen to have progressed through the sequence outlined above.

3.2 Origins of Site Catchment Analysis

The concept of site catchment analysis was first proposed by Vita-Finzi and Higgs (1970) as the “...*study of the relationships between technology and those natural resources lying within economic range of individual sites.*”. Their process of site catchment analysis, as first presented, has been summarised by Bailey and Davidson (1983, 89) as:

1. defining a problem in the archaeological literature at the time, with a proposed methodological solution.
2. drawing upon ethnographic work and fundamental energetics (e.g. Chisholm 1968 and Lee 1969), stating as fact that there are always some spatial limitations to the field of potential resources to which the inhabitants of a given community had access.
3. identifying several resource categories (i.e. arable land, grazing land) and assigning to them variables relevant to the investigation of site exploitation territories.

The very basis of site catchment analysis relies on the assumption that “... the further the area is from the site, the less it is likely to be exploited ...” (Vita-Finzi and Higgs 1970, 7)

In nearly all academic papers regarding site catchment analysis, two references are consistently used to support the above assumption: *!Kung Bushman Subsistence: An Input-Output Analysis* by Lee (1969) and *Rural Settlement and Land Use* by Chisholm (1968). Lee, in an ethnographic study of the Kalahari Bushman hunter-gatherers, finds that women do not generally forage for food (plants) beyond 10 km from their main camp, and although the men sometimes travel further when hunting, they generally

set up a temporary over-night camp when they do. Chisholm, on the other hand, finds that there is a general 3-4 km limit between farmers and their fields (Chisholm 1968, 62;125;146;147), noting that only exceptionally do farmers travel further, and that “at a distance of 3-4 kilometres the costs of cultivation necessitate a *radical* modification of the system of cultivation or settlement – for example by the establishment of subsidiary settlements – though adjustments are apparent before this point is reached.” (Chisholm 1968, 148). It was studies like these that prompted Vita-Finzi and Higgs to use what seems to have become the now de-facto standard catchment radius of 5 km. The reasoning behind this was that the resources the community would have required would have come from the 7854 ha ($\pi \cdot 5^2 = 78.5398163 \text{ km}^2$) catchment that the 5 km limit encompassed which the authors took as “... the economic threshold for the more intensive subsistence agricultural economies.” (Vita-Finzi and Higgs 1970, 16). By examining the land in the now ascribed catchment area, they were able to classify the land into the following categories: arable, potentially arable, grazing, marsh and dunes.

Each category was then expressed in terms of percent of the total catchment. By examining the ratio of the weight adjusted land types, they drew conclusions regarding site function (e.g. if a site’s catchment area contained low percentages of arable land the settlement’s subsistence strategy was deemed to be largely non-agricultural) (Flannery 1976a, 92). It is important, however, to draw attention to the fact that 20% of 7854 ha is 1570.8 ha; problematic because 1570 ha was more land than was required, a fact which was not lost to critics of their original methodology.

The original methodology for site catchment analysis by Vita-Finzi and Higgs (1970) included a rudimentary method for dealing with travel time. Vita-Finzi and Higgs recognised that walking time was an important factor for determining the site catchment boundary. They did not directly apply this to agri-

Distance in kilometres	Net Output (after Virri 1946, 7)	Arbitrary Weighting adopted (as percent)
0-0.1	100	–
0.5	67	–
1.0	50	100
1.5	40	–
2.0	33	50
3.0	25	33
4.0	20	25
5.0	17	20

Table 3.1: Weightings for agricultural catchment bands (after Vita-Finzi and Higgs 1970, Table 3)

cultural catchments but instead drew on work done by Virri (Vita-Finzi and Higgs 1970, 29), who found that, in Finland farmsteads, both net and gross yields per hectare decreased with the distance from the farmstead. To compensate for the exclusion of walking time in their methodology for agricultural land catchments, they instead implemented a system of weighted concentric circles drawn around sites. This weighting factor was applied at increments of 1 km starting at the settlement and stopping when the radius reached 5 km. The categorised land contained within each band formed by

the concentric circles was then given a weighting (effectively this is of diminishing productivity), and the potential contributions from the resource contained in each band was then reduced accordingly. The weighting factors are shown in Table 3.1.

There is a potential problem with the applicability of the work Virri did that, to my knowledge, has not been satisfactorily addressed^{3.1}. The issue I have with the progressively diminished agricultural productivity is that Virri (1946) was talking about *farmsteads* and not towns or cities. Of course it makes sense that a farmer will be unable to give as much attention to fields which are further away from his home in comparison to those that are close; applying this to ancient settlements where hundreds and thousands of people lived together in a settlement, however, is in my opinion a completely different thing. The person who is tending their crop will have to walk from their residence in the settlement (where they reside with everyone else) to and from their land, and unless every farmer in the settlement had an incredibly narrow pie shaped wedge of land originating at the settlement and consistently expanding directly outwards, this methodology should not be applied. On the contrary, I would argue that as the distance from the settlement increased, productivity might actually increase. This position is based on the principle that cultivation of land would occur on an as needed bases, therefore the 'newest' cultivated land would most likely have been that land further from the settlement, as it would have been in use for less time than the land immediately surrounding the site, thus being more fertile and less depleted. At any rate, until there is evidence of ancient farmers living in the centre of their fields, the methods for catchment analysis proposed in this thesis seem to be a much more suitable approach than that of progressively diminished agricultural productivity based on distance from a settlement.

Site catchment analysis has been continually evolving with various refinements, enhancements and additions since its conception. One important development occurred in 1976 when Flannery (1976a) presented additional variants of site catchment analysis. After initially presenting the San Lorenzo case study (contributed by Rossmann 1976, 95) using an implementation of the original format pioneered by Vita-Finzi and Higgs, Flannery himself presented a section on Oaxaca and Tehuacán where he reversed the process by examining the empirical data of the resources found at the site and determining where they came from. Using this approach, he avoided the problem of defining site function based on landscape, which was one of the often repeated criticisms of the Vita-Finzi and Higgs methodology (for examples see Roper 1979, R.W. Chapman in Clarke 1978, 125-128, Hodder and Orton 1976 and Wobst 1978). This is

^{3.1}The Virri (1946) paper was difficult to get. Only in the final 3 weeks before submitting this thesis was I able to locate it, thanks to the help of Anu Riikonen who was able to finally locate it in the University of Helsinki Library archives. Further thanks to Anu for his help with the translation of the paper. Finally, one reason for the difficulty of finding this paper was due to the fact that previous bibliography entries have been (consistently) misspelled, without a Volume number, and with an incorrect journal name as well.

important and directly relevant to this research because it is this ‘what is here, and how can we use it?’ approach which has largely been taken throughout this thesis.

For further examples of early site catchment analysis methodologies, the reader can be directed to several sources (see Table 3.2).

Table 3.2: Early publications focussing on different catchment analysis methodologies.

Methodological Focus	Reference
Weighting of sites	Dennel and Webley (1975) Peebles (1978) Webley (1972)
Off/on site comparison quantiles	Akazawa (1980) Bailey (1978) Bailey (1982) Davidson (1976) Hastorf (1980) Jarman et al. (1976) Munson and Yarnell (1971)
Predictive modelling of site locations	Bailey (1978) Gamble (1979) Jochim (1976) Schwellnus (1974) Sturdy (1975)
Analysis of locations relative to each other checking assumptions about territory size and site function	Brumfiel (1976) Cassels (1972) Dennel and Webley (1975)
Regional analysis as a check on territorial site data interpretation	Bay-Petersen (1982) Gamble (1978) Sturdy (1975) Zarky (1976)

Bailey and Davidson (1983, 88) define six objectives of site catchment analysis as:

1. defining areas regularly used by the inhabitants of a settlement for their day to day subsistence
2. tracing materials and resources present in the archaeological assemblage to their points of origin in the surrounding landscape
3. reconstruction of micro-environments surrounding a settlement to gain insight into variations of palaeoenvironmental data evident in the archaeological record.
4. reconstructing the potential food resources which would have been available to the inhabitants of the settlement, which infers the subsistence economy they practised.

5. reconstructing the possible function of settlements
6. reconstructing socio-economic ties between settlements when considering them as members of regional settlement systems

It has been important to outline the origins of site catchment analysis as a methodological/procedural tool because it shows how the process was limited by technological shortcomings. These shortcomings have been vastly reduced by developments in both hardware and software, and the following section describes how GIS has become an indispensable tool in site catchment analysis.

3.3 Geographic Information Science

Today's commonly used technique of utilising aerial photography and satellite imagery for archaeological investigation began in 1922 (Olsen 1985). It is now pertinent to examine how remote sensing in archaeology spawned a reactionary process which resulted in the development of one of archaeology's most powerful analytical tools: the modern day Geographical Information System.

3.3.1 Remote Sensing

Remote sensing made it possible to survey areas much more quickly. Remote sensing also made it possible to conduct surveys in areas where the landscape was difficult or impossible to traverse because it could now be done without being physically on the ground. The resultant images provided an efficient way of storing/recording vast amounts of data for large tracts of land. Countless other advancements in technology also contributed to this burgeoning pool of data from which archaeologists could draw. The problem was, however, that there were no effective tools for working with this vast amount of data which meant that remote sensing could only make a limited contribution to archaeological knowledge; with the advent of computers, archaeologists were given the tool that enabled them to finally make use of the available data.

One tool particularly effective for working with high levels of spatial data in a fast, systematic and efficient manner is the Geographic Information System (henceforth GIS) which allows for a rapid and systematic way to combine spatial data from a wide number of different sources. Combining spatial data was first illustrated in 1950 when Jacqueline Tyrwhitt took data regarding elevation, hydrology, geology, and cropland and combined them into a single map (Tyrwhitt 1950).

3.3.2 Map Overlays

The pioneering work done by Tyrwhitt was done entirely by hand, literally cutting sections from one map and lining them up on another, thus superimposing the cut-outs

as ‘map layers’. Nine years later, this task was able to be done with a computer thanks to Tobler who developed MIMO (Map In-Map Out). MIMO used a three step process: map input, map manipulation, and map output (Tobler 1959). Although not powerful by today’s standards, this pioneering effort served to simplify the process of map overlays. As map overlays became easier and more common, it triggered further development and resulted in, among other things, the United States Central Intelligence Agency’s *World Data Bank* and the US Census Bureau’s *Dual Independent Map Encoding* (DIME); a revolutionary system for encoding digital maps that is still in use today (Clarke 1997, 8). The rapidly increasing volume of digital data spawned the development of tools to deal with it. One of the first GIS programs developed was the Canada Geographical Information System, or CGIS, introduced in 1964.

3.3.3 Defining GIS

Fundamentally, a GIS is a tool which allows for the manipulation of spatial data. However, a GIS provides further analytical possibilities. Because of the highly adaptable nature of a GIS, it is difficult to provide a single explicit definition. An example of a simple definition is:

“A powerful set of tools for storing and retrieving at will, transforming and displaying spatial data from the real world for a particular set of purposes”

– Burrough (1986, 6).

The above definition provides no clues as to what the purpose of a GIS could be, nor does it explain how the data is represented or stored. This definition is contrasted with:

“A geographic information system (GIS) is an information system that is designed to work with data referenced by spatial or geographic coordinates. In other words, a GIS is both a database system with specific capabilities for spatially-referenced data, as well [as] a set of operations for working with data . . . In a sense, a GIS may be thought of as a higher-order map.”

– Star and Estes (1990, 2-3).

Star and Estes provides a more explicit definition including a more precise definition of the type of data a GIS can work with. A more succinct definition is:

“A geographic information system is a special case of information systems where the database consists of observations on spatially distributed features, activities or events, which are definable in space as points, lines, or areas. A geographic information system manipulates data about these points, lines and areas to retrieve data for ad hoc queries and analyses”

– Duecker (1979, 106).

Duecker addresses how data is represented, and expands on the type of data suitable for inclusion in a GIS, but fails to convey the broader potentials of a GIS (e.g. modelling).

An important component of any GIS is its database, but a database in a GIS is different from a 'normal' database: it is 'spatially aware' (capable of determining spatial relationships with data). For a database to be spatially enabled, it must incorporate (Wheatley and Gillings 2002, 23) a:

- **locational component:** the 'position' in terms of geographical space and determines where (and in what form) things are.
- **topological component:** referring to the logical relationships between the various objects in the geographic database.
- **attribute component:** those features (or characteristics) of objects that determine what things represent and their properties.
- **metadata component:** the detailed documentation providing all relevant information pertaining to each object in the spatially enabled database

A GIS can combine different types of spatially-referenced data that can result in new outcomes, meaning that a GIS can also be described as a tool capable of producing new information. The new information provided by GIS can contribute knowledge that was previously unobtainable.

3.3.4 GIS in Archaeology

The use of the term 'GIS' began appearing in archaeological publications between 1983 and 1985 (Kvamme 1996). As the focus in archaeology began shifting from the settlement to the landscape (Warren 1990a), and data was becoming more readily available for immediate analysis, there was a growing need for automated predictive modelling. Predictive modelling establishes a 'causal' relationship between archaeological site locations and various environmental factors upon which a statistical model is based (Gourad 1999). Computers became the predominant tool for creating the statistical models that become a convention in archaeology (Aldenderfer 1987). GIS is a very effective tool to use for predictive modelling, and it is perhaps this feature of a GIS which more than anything made GIS a mainstream tool in archaeology (Gourad 1999). With the introduction of relational databases, archaeology as a discipline was impacted by changing the way that data was collected (e.g. remote sensing), the nature of questions regarding the data (e.g. the ability to explore complex logistical regression models), and, as some have suggested, even the general conceptual design of an archaeological dig (Harris and Lock 1996, 352).

3.4 Modelling

Sebastian and Judge (1988) suggested that predictive modelling became increasingly important as archaeology went from a largely descriptive stage to a more explanatory stage.

Drawing on the mathematical modelling undertaken by Orton (1980, 20), Voorrips (1987, 61) defines archaeological models as “... [the] partial representations of a theory ... formulated in a manner which enables the archaeologist to test the theory by means of empirical data”. Warren describes predictions as a critical component of scientific method; hypotheses are formed to explain new patterns found in various contexts, and models are then built in different contexts in order to test the hypothesis (Warren 1990b).

Regression analysis is a statistical technique for determining the relationships that exist between variables. There are several kinds of regression analysis (e.g. linear, non-linear, stepwise) and predictive modelling in archaeology draws upon all of them. When using a two variable method (bivariate regression^{3.2}), one variable is known as an ‘independent’ variable (e.g. proximity to a water source) and the other variable is a ‘dependent’ variable (e.g. number of sites). A value is *chosen* for the independent variable, and the value for the dependent variable is then calculated; for example, if determining how many sites (*s*) are located within *x* km of spring, the independent variable is the distance from (or proximity to) a spring (*x*), and the dependent variable is the number of sites found (*s*). When the complexity of a model increases, and more variables are introduced, it is called multivariate regression. Multivariate regression models look at the effects of one independent variable on multiple dependant variables. The next level of regression modelling becomes much more complex; logistical regression models examine the relationships between **multiple independent** variables and **multiple dependent** variables. To make this even more complex, each independent variable can have as many parameters as the archaeologist wants (Savage 1990). *Step-wise* logistical regression is the extension of logistical regression, different only because it uses pre-selected significance levels where variables get added or removed (Christopher et al. 1996). The methods best suited to archaeological applications are the two latter (and most complicated) methods and also describes the type of modelling used throughout this thesis. The above discussion describe a few select examples highlighting some of the early applications of computers in archaeology.

The examples discussed above are drawn from published work that is replete with diverse examples that utilise the continually increasing power of both computer hardware and software (e.g. Bevan 2008, Fletcher 2008, Kvamme 2008, Fitzjohn 2007, Wilkinson et al. 2007, Kvamme 2007, Given et al. 2007, Ebert 2006, Hare 2004, Baja

^{3.2}Regression is defined as “a measure of the relation between the mean value of one variable (e.g. output) and corresponding values of other variables (e.g. time and cost).” – Jewell et al. (2001)

et al. 2001, Ebert 2000).

3.4.1 Modelling Landuse in GIS

In the last 7 years, there have been a number of interesting attempts to model past land use and landscape evolution in a GIS environment. In particular, the work of Robb and Van Hove, Ayala and French, Wainwright and Legg and Taylor has been particularly influential in the design and implementation of the methodology in this research.

Ayala and French examined landscape development of north-central Sicily, integrating geomorphological research of the Fiume di Sotto di Troina valley with archaeological survey in a GIS to model past landuse practices of the Roman and Medieval periods and the possible potential erosion caused by different land use practices throughout history (Ayala and French 2005). An important outcome from this work was that their erosion models showed “[how intensive geoarchaeological survey]... can aid in the interpretation of occupational history in a way that neither geomorphological nor archaeological investigation can accomplish on their own.” (Ayala and French 2005, 165). The ability of Landuse Analyst (see Section 4.3 on page 76) to more accurately and systematically model a settlement’s specific catchment area and land use strategies would have been a valuable tool for inclusion in their methodology, and in fact is an interesting potential future collaborative project which could be explored.

Legg and Taylor examined the influence of environmental factors on the distribution Irish ringforts in the Inny River catchment area. Along with environmental variables, locational information was entered into a GIS, and a model was created for predicting the density of ringforts. When extended outside of the Inny River catchment area, their model succeeded in producing results that were considered broadly consistent with recorded site distributions (Legg and Taylor 2006, 201,214–215). The methodology developed in this thesis for both the network and site catchment analysis would provide an interesting way to look at how the distribution of these ringforts could also have been linked to subsistence strategies, and in combination with their work, possibly result in a more robust and accurate model^{3.3}.

In his 2008 paper, Wainwright (a geographer at the University of Sheffield) uses an agent based model to help understand human-landscape interactions. Basic actions like the acquisition of food are simulated to the scale of the individual. The CybErosion modelling framework he presents attempts to include human actions into a landform evolution model by simulating how people interacted with their landscape (Wainwright 2008, 659). It would be very interesting to incorporate this level of human modelling into the methodology and software developed for the research presented in this thesis, and in doing so, give the agent based model a further platform by which it could include many more features provided by Landuse Analyst, such as crop/herd/dairy modelling

^{3.3}This potential is pointed out in the conclusions of this thesis in Chapter 7, page 360

and regional network analysis.

Perhaps the most pertinent piece of research for my research in the last 7 years, however, has been carried out on Neolithic Calabria in southern Italy by Van Hove (e.g. Van Hove 2002, 2003a,b, Robb and Van Hove 2003).

Robb and Van Hove (2003) explored the relationships between Neolithic land use and culture in southern Italy. In this paper, they focus on how cultural choices affected land use and the resulting human landscape. GIS was used to "...trace out the implications of this cultural choice systematically." (Robb and Van Hove 2003, 242). They attempted to reconstruct land use practices and what the resulting landscape would have looked like by identifying the specific areas which would have been utilised for different agricultural activities (e.g. herding/foraging, farming). They did this in an effort to provide support for contrasting theories of the Neolithic economy and culture.

Similar to the approach taken in this thesis, reasonable estimates of many input factors were used, and the effects of varying these values observed in lieu of trying to determine one single 'perfect' model (Robb and Van Hove 2003, 245). Land classification was done using zones derived from topographical maps, and further classified with a heavy reliance on slope (Robb and Van Hove 2003, Figure 2, page 246). What this study failed to do, however, was to provide a quantifiable method, based on archaeological evidence, to calculate food production and land requirements. Instead, land area requirements were based upon estimations of Neolithic agriculture: "The next step was to estimate the resources a group of 50 people would have required annually to survive. A family of 4-6 people living primarily on grain would have required about 1000kg per year ... and Neolithic grain yields are thought to have been between 500 and 1000 kg/ha ... [so] a group of 50 would have required perhaps 10-15 ha of active grain plots with an additional amount in fallow fields and gardens of other crops." (Robb and Van Hove 2003, 246).

In order to locate the specific areas of probable farming land a simple slope based cost surface ($friction = \log(slope + 1) + .5$) was used to calculate travel effort. Further, 20 permutations of the model were examined (Robb and Van Hove 2003, 247). They were able to conclude that "The most important point of our reconstruction of Neolithic land use is that arable land was probably never the limiting resource on Neolithic settlement..." (Robb and Van Hove 2003, 247).

Significantly, these conclusions were based on unquantifiable estimates^{3,4} of many of their input factors. In addition, their model is limited by the fact that a static number of 50 was used to model population, and that their 20 permutations were effectively based on three input variables (percentage levels of crops, herds and wild-resources). More significantly, the 20 permutations were determined using values which they felt fit the description assigned to the permutation. For example (Robb and Van Hove 2003,

^{3,4}I wish only to point out that these estimates are unquantifiable; they were all well referenced

Table 1, page 248) labels the 20 Models (permutations) as Foragers, Forager-Farmer, Farmer, Pastoralist and one as Mixed. A more comprehensive model, like that developed in this thesis, would have proved a much more meaningful and robust method to calculate the total land needs. It would also avoid the problem of assigning input values based on what we feel suit a certain strategy; rather, the model outcome could be analysed, and the labelling could be done based on this. In other words the strategies would be defined by the data, not by the pre-conceived notion of what certain strategies are.

The methodology and software developed for and implemented in this thesis attempts to directly address the problems that using unquantifiable input variables presents to our understanding of past land use. Specifically, algorithms have been developed for the calculation of many of the input variables which in previous studies were simply estimated. The remainder of this thesis is dedicated to the presentation and implementation of this methodology.

3.5 Conclusions

In this chapter, the origins of site catchment analysis and the basics of computer modelling in archaeology have been discussed. Site catchment analysis, as first proposed for use in site catchment analysis by Vita-Finzi and Higgs (1970), has been employed with few modifications by researchers right up to the present day (e.g. Campbell 2009, 111). The reader will see, in the following chapter where the methodology of this research is presented, how the use of GIS and logistical regression analysis modelling have been combined into a versatile tool which provides a fast, systematic and improved approach to modelling subsistence catchment areas. The effectiveness of this tool will be evident in Chapters 6 and 7, where it is employed to analyse Chalcolithic and EBA Tel esh Shuna North and its surrounding region.